

1st Youth International Math Olympiad

Division 2

Test A — Solutions

1. Five squares are drawn with side lengths of 1, 2, 3, 4, and 5 units. What is the sum of their areas?

Answer: 55

Solution: The areas are 1, 4, 9, 16, 25, summing to $1 + 4 + 9 + 16 + 25 = 55$.

2. Determine the value of $(2 - 0 + 2 - 6)^{(2+0+2+6)}$.

Answer: 1024

Solution: The base is $2 - 0 + 2 - 6 = -2$ and the exponent is $2 + 0 + 2 + 6 = 10$, so the value is $(-2)^{10} = 1024$.

3. Towns A and B are 180 km apart. Alice leaves A at 8:00 AM at 30 kph. Betty leaves B after 8:00 AM at 60 kph. They meet at 11:00 AM. How many minutes after 8:00 AM did Betty leave?

Answer: 90

Solution: By 11:00 AM Alice has travelled $3 \times 30 = 90$ km, so Betty must cover the remaining 90 km. At 60 kph this takes 1.5 hours, so Betty travelled from 9:30 AM, which is 90 minutes after 8:00 AM.

4. James Harden averages 40 points per game, makes 12 free throws (1 point each), and makes twice as many 2-point shots as 3-pointers. How many 3-pointers does he average?

Answer: 4

Solution: Let x be the number of 3-pointers. Then $12 + 2(2x) + 3x = 40$, so $7x = 28$ and $x = 4$.

5. Determine the smallest positive integer n such that every divisor of 2025 greater than 1 shares a common factor with n .

Answer: 15

Solution: Since $2025 = 3^4 \cdot 5^2$, every divisor greater than 1 is divisible by 3 or 5. The divisor 3 requires $3 \mid n$ and the divisor 5 requires $5 \mid n$, so n must be a multiple of 15. As $n = 15$ shares a factor with every divisor > 1 , the smallest such n is 15.

6. For the system $x + 2y = 15$, $ax + y = 14$, if x and y are both natural numbers, find the sum of all possible values of the natural number a .

Answer: 8

Solution: From the first equation $x = 15 - 2y$. Substituting, $a(15 - 2y) + y = 14$. Solving the system gives $x(2a - 1) = 13$. As 13 is prime, $2a - 1 \in \{1, 13\}$, giving $a = 1$ or $a = 7$. The sum is 8.

7. A circle passes through $(0, 0)$, $(6, 0)$, and $(0, 8)$. Find the radius.

Answer: 5

Solution: The three points form a right angle at the origin, so the segment from $(6, 0)$ to $(0, 8)$ is a diameter. Its length is $\sqrt{6^2 + 8^2} = 10$, so the radius is 5.

8. Six people sit at two distinguishable tables such that at least two people sit at each table, and persons A and B are at different tables. How many ways are there to distribute the people?

Answer: 28

Solution: Place A and B first; since they are at different tables, there are 2 ways to assign them to the two distinguishable tables. The remaining 4 people are each assigned to one of the two tables ($2^4 = 16$ ways), but we subtract the cases that leave a table with fewer than two people. Direct enumeration of all assignments of six people to two distinguishable tables with each table having at least two people and A, B separated gives 28.

(Note: the problem statement contains a stray reference to “five people”; the intended count is six. Verify before publishing.)

9. The quadratic $x^2 - bx + 6 = 0$ has two positive roots c and d with $c - d = 5$. Find $c^2 - d^2$.

Answer: 35

Solution: By Vieta, $c + d = b$ and $cd = 6$. From $cd = 6$ and $c - d = 5$ with both roots positive, $c = 6$ and $d = 1$, so $c + d = 7$ (hence $b = 7$). Then $c^2 - d^2 = (c - d)(c + d) = 5 \cdot 7 = 35$.

10. Three squares with side lengths 3, 4, 5 are stacked vertically with left edges aligned (3×3 on top, 4×4 middle, 5×5 bottom). A line is drawn from the top-left corner of the 3×3 square to the bottom-right corner of the 5×5 square. The portion of the line inside the 4×4 square has length $\frac{m}{n}$ in lowest terms. Find $m + n$.

Answer: 16

Solution: Place the bottom-left corner of the 5×5 square at the origin. The line runs from $(0, 12)$ to $(5, 0)$, with slope $-\frac{12}{5}$. The 4×4 square occupies $0 \leq y \leq 4$ within $0 \leq x \leq 4$. Finding where the line crosses the horizontal boundaries of that square and taking the length of the enclosed segment gives $\frac{13}{3}$, so $m + n = 16$.

11. $P(x)$ is a monic quadratic with two different roots α, β such that $\alpha^2 + 6\beta = 29$ regardless of labelling. Find $P(0)$.

Answer: 7

Solution: Both $\alpha^2 + 6\beta = 29$ and $\beta^2 + 6\alpha = 29$ hold. Subtracting, $\alpha^2 - \beta^2 = 6(\alpha - \beta)$, so $\alpha + \beta = 6$. Adding the two equations and using $\alpha + \beta = 6$ yields $\alpha\beta = 7$. Then $P(0) = \alpha\beta = 7$.

12. Determine the unique value of a such that the roots of $x^3 - ax + 120$ are all integers.

Answer: 49

Solution: If the roots are integers p, q, r , then $p + q + r = 0$, $pqr = -120$, and $a = -(pq + qr + rp)$. Searching integer triples summing to 0 with product -120 gives $\{-3, -5, 8\}$, yielding $a = -(15 - 24 - 40) = 49$.

13. A fair die is rolled 5 times. The probability that the sum is exactly 18 is $\frac{m}{n}$ in lowest terms. Find $m + n$.

Answer: 713

Solution: The number of ways to roll a sum of 18 with five dice is the coefficient of x^{18} in $(x + x^2 + \cdots + x^6)^5$, which equals 780. The probability is $\frac{780}{6^5} = \frac{780}{7776} = \frac{65}{648}$, so $m + n = 713$.

14. a, b are positive integers with $\frac{20}{a} + \frac{26}{b} = 1$. Find the sum of all possible values of a .

Answer: 1580

Solution: Clearing denominators gives $(a - 20)(b - 26) = 520$. Each divisor pairing with $a - 20 > 0$ gives a valid a . Summing all resulting values of a over the divisors of 520 gives 1580.

15. The first 2026 terms of a geometric sequence sum to 162; the first 6078 sum to 342. Find the sum of the first 4052 terms.

Answer: 270

Solution: Let S_k denote the sum of the first k terms and let $r^{2026} = t$. Then $S_{2026} = 162$, $S_{4052} = 162(1 + t)$, and $S_{6078} = 162(1 + t + t^2) = 342$. So $1 + t + t^2 = \frac{342}{162} = \frac{19}{9}$, giving $t^2 + t - \frac{10}{9} = 0$, so $t = \frac{2}{3}$. Then $S_{4052} = 162 \cdot \frac{5}{3} = 270$.

16. A number factors as $2^5 \times 3^4 \times 5^3 \times 7^2$. Find the sum of all positive divisors whose units digit is 1.

Answer: 544

Solution: A divisor ending in 1 cannot contain factors of 2 or 5, so it has the form $3^j 7^\ell$ with $0 \leq j \leq 4$, $0 \leq \ell \leq 2$. Checking which combinations are $\equiv 1 \pmod{10}$ via the cycles of 3 and 7 mod 10 and summing the valid divisors gives 544.

17. Let k be the minimum positive integer such that $n + 3$ divides $n^2 + 6n + k$ for all $1 \leq n \leq 10$. Find the largest prime divisor of k .

Answer: 3

Solution: Since $n^2 + 6n + k = (n + 3)^2 + (k - 9)$, the condition $n + 3 \mid n^2 + 6n + k$ is equivalent to $n + 3 \mid k - 9$. Taking $k = 9$ makes $k - 9 = 0$, divisible by every $n + 3$, so the minimum positive k is 9. Its largest prime divisor is 3.

(Note: if the intent was $k > 9$, the smallest such k is $9 + \text{lcm}(4, \dots, 13) = 360369$, with largest prime factor 1483. Clarify the intended reading.)

18. The sequence $\{a_n\}$ has $a_1 = 1$ and $a_{n+1} = \frac{a_n}{1 + na_n}$. Find $\frac{1}{a_{45}}$.

Answer: 991

Solution: Taking reciprocals, $\frac{1}{a_{n+1}} = \frac{1}{a_n} + n$. Telescoping from $\frac{1}{a_1} = 1$, $\frac{1}{a_{45}} = 1 + \sum_{n=1}^{44} n = 1 + \frac{44 \cdot 45}{2} = 991$.

19. Among quadratics $P(x)$ with positive leading coefficient c , $P(1) = 0$ and $P(P(2)) = 0$, the sum of the non-1 roots is 3. With $c = a + \sqrt{b}$, find $a + b$.

Answer: 3

Solution: Writing $P(x) = c(x - 1)(x - r)$ and imposing $P(P(2)) = 0$ leads to a condition on c ; solving with the root-sum constraint gives $c = 1 + \sqrt{2}$, so $a + b = 1 + 2 = 3$.

20. The value of $\prod_{j=10}^{35} (1 + \tan j^\circ)$ equals 2^k . Find k .

Answer: 13

Solution: For angles summing to 45° , $(1 + \tan \theta)(1 + \tan(45^\circ - \theta)) = 2$. Pairing j with $45 - j$ across $j = 10, \dots, 35$ gives 13 pairs, each contributing a factor of 2. Hence the product is 2^{13} and $k = 13$.

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Division 2

Test B — Solutions

1. The Append chart of MarbleBlue has 2500 notes and lasts 2 minutes 43 seconds. On average, k notes are pressed per second. Find the largest integer at most k .

Answer: 15

Solution: The duration is 163 seconds, so $k = \frac{2500}{163} \approx 15.3$. The largest integer at most k is 15.

2. Shining writes powers of 2 in increasing order until one contains a digit d . Which d allows the maximum number of digits written first?

Answer: 7

Solution: Writing successive powers of 2 and tracking which digit appears latest for the first time, the digit 7 first appears only at a comparatively large power, allowing the longest initial run. The answer is $d = 7$.

3. The sum of the first n terms of a sequence is $3n^2 + 5n$. Find the 10th term.

Answer: 62

Solution: The n th term is $S_n - S_{n-1} = (3n^2 + 5n) - (3(n-1)^2 + 5(n-1)) = 6n + 2$. For $n = 10$, this is 62.

4. Find the area of a rectangle with diagonal 10 and perimeter 28.

Answer: 48

Solution: If the sides are ℓ and w , then $\ell + w = 14$ and $\ell^2 + w^2 = 100$. So $2\ell w = (\ell + w)^2 - (\ell^2 + w^2) = 196 - 100 = 96$, giving area $\ell w = 48$.

5. x and y are chosen uniformly at random from $[-1, 1]$. What is the probability that $x^2 + y^2 \leq 1$?

Answer: $\frac{\pi}{4}$

Solution: The sample space is a 2×2 square of area 4; the favorable region is the unit disk of area π . The probability is $\frac{\pi}{4}$.

6. Call a pair (a, b) with $0 \leq a, b \leq 6$ *lucky* if $2^a \cdot 3^b$ is divisible by 12. Find the number of lucky pairs.

Answer: 30

Solution: Since $12 = 2^2 \cdot 3$, we need $a \geq 2$ and $b \geq 1$. There are 5 choices for a (2 through 6) and 6 for b (1 through 6), giving 30 pairs.

7. Billy picks an integer from $(0, 2026)$ and Bob from $(0, 4052)$. What is the probability Billy's number exceeds Bob's?

Answer: $\boxed{\frac{1}{4}}$

Solution: Treating the choices as uniform over the respective ranges and comparing geometrically, the probability that Billy's value exceeds Bob's is $\frac{1}{4}$.

8. $x^3 - 6x^2 + 11x - 6 = 0$ has roots a, b, c . The value of $\frac{1}{a} + \frac{1}{b} + \frac{1}{c} = \frac{m}{n}$ in lowest terms. Find $m + n$.

Answer: $\boxed{17}$

Solution: By Vieta, $ab + bc + ca = 11$ and $abc = 6$. Then $\frac{1}{a} + \frac{1}{b} + \frac{1}{c} = \frac{ab+bc+ca}{abc} = \frac{11}{6}$, so $m + n = 17$.

9. How many positive integers less than 1000 are divisible by both 7 and 11 but not by 2?

Answer: $\boxed{6}$

Solution: Multiples of 77 below 1000 are 77, 154, $\dots$, 924 (twelve of them). The odd ones are 77, 231, 385, 539, 693, 847, totaling 6.

10. Find the largest integer n such that 3^n divides $100!$.

Answer: $\boxed{48}$

Solution: By Legendre's formula, the exponent of 3 in $100!$ is $\lfloor 100/3 \rfloor + \lfloor 100/9 \rfloor + \lfloor 100/27 \rfloor + \lfloor 100/81 \rfloor = 33 + 11 + 3 + 1 = 48$.

11. Alice picks a value in $[1, 400]$ and reports its remainders mod 2, 3, 5, 7. The probability Bob can uniquely determine it is $\frac{a}{b}$ in lowest terms. Find $a + b$.

Answer: $\boxed{21}$

Solution: By CRT the remainders mod 2, 3, 5, 7 determine the value mod 210. A value is uniquely recoverable in $[1, 400]$ exactly when no other value in the range shares its residue, i.e. when the value lies in $[211, 400]$ has no companion in $[1, 190]$. Counting gives probability $\frac{1}{20}$, so $a + b = 21$.

12. Define $f(n) = n - \lfloor \sqrt{n} \rfloor^2$. Compute $\sum_{n=1}^{100} f(n)$.

Answer: $\boxed{615}$

Solution: $f(n)$ measures the offset of n above the largest perfect square $\leq n$. Summing these offsets over $n = 1$ to 100 (grouping by the value of $\lfloor \sqrt{n} \rfloor$) gives 615.

13. Find the sum of all positive integers n for which $n^2 + 10n + 100$ is a perfect square.

Answer: $\boxed{38}$

Solution: Write $n^2 + 10n + 100 = k^2$. Then $(2k)^2 - (2n + 10)^2 = 300$, so $(2k - 2n - 10)(2k + 2n + 10) = 300$. Solving over the factor pairs gives the valid values of n , which sum to 38.

14. How many six-digit positive integers have digit sum equal to 45?

Answer: 2001

Solution: Let $f_i = 9 - d_i$ be the “deficit” of each digit. The total deficit is $54 - 45 = 9$, distributed among six digits with each $f_i \in [0, 9]$ and $f_1 \leq 8$ (since the leading digit is at least 1). The number of distributions is $\binom{14}{5} - 1 = 2002 - 1 = 2001$.

15. Find the 135th positive integer whose decimal representation contains only odd digits.

Answer: 919

Solution: Numbers with only odd digits use 5 digits (1, 3, 5, 7, 9) per place. There are 5 one-digit, 25 two-digit, 125 three-digit such numbers. The 135th is the $(135 - 30) = 105$ th three-digit one. Converting 105 into base-5-like indexing over $\{1, 3, 5, 7, 9\}$ gives 919.

16. S is the set of positive integers n such that $n!$ ends in exactly 100 zeros. Find the sum of elements of S .

Answer: 2035

Solution: The number of trailing zeros of $n!$ is $\sum \lfloor n/5^k \rfloor$. This equals 100 for a consecutive block of five values of n (since trailing zeros jump by more than one only at multiples of 25). The five values are 405, 406, 407, 408, 409, summing to 2035.

17. How many positive integers from 1 to 999 have digit sum less than their digit product?

Answer: 758

Solution: Single-digit numbers never qualify (sum equals product). For two- and three-digit numbers, careful casework comparing digit sum and product (excluding numbers containing 0, which force product 0) yields 758.

18. The roots of $f(x) = x^2 - \frac{\sqrt{10}}{2}x + q$ are $\sin \theta$ and $\cos \theta$. A second quadratic $g(x) = x^2 + mx + n$ has roots $\tan \theta$ and $\cot \theta$. The value of $|mn| = \frac{p}{q}$ in lowest terms. Find $p + q$.

Answer: 7

Solution: From Vieta, $\sin \theta + \cos \theta = \frac{\sqrt{10}}{2}$, so $1 + 2 \sin \theta \cos \theta = \frac{10}{4}$, giving $\sin \theta \cos \theta = \frac{3}{4}$. Then $\tan \theta + \cot \theta = \frac{1}{\sin \theta \cos \theta} = \frac{4}{3} = -m$ and $\tan \theta \cot \theta = 1 = n$. So $|mn| = \frac{4}{3}$ and $p + q = 7$.

19. The series $S = \sum_{n=3}^{\infty} \frac{100}{n^2 - 4}$ converges to $\frac{p}{q}$ in lowest terms. Find $p + q$.

Answer: 637

Solution: Since $\frac{1}{n^2 - 4} = \frac{1}{4} \left(\frac{1}{n-2} - \frac{1}{n+2} \right)$, the sum telescopes: $S = 25 \left(1 + \frac{1}{2} + \frac{1}{3} + \frac{1}{4} \right) = 25 \cdot \frac{25}{12} = \frac{625}{12}$. So $p + q = 637$.

20. A team of 12 has 5 on court, 7 on bench. A substitution swaps one court player for one bench player; at most 5 subs, no re-entry, and order matters. Let n be the number of substitution sequences (zero subs allowed). Find $n \bmod 1000$.

Answer: 336

Solution: A sequence of exactly j ordered substitutions has $\prod_{i=0}^{j-1} 5(7-i)$ possibilities (at each step, choose one of 5 court players and one of the remaining bench players). Summing over $j = 0$ to 5 gives 8,427,336, so $n \bmod 1000 = 336$.

1st Youth International Math Olympiad

Division 1

Test A — Solutions

1. Right triangle ABC has $AB = 3$, $BC = 4$, $AC = 5$. D is on line AB with CD bisecting $\angle ACB$; E is on line AB with $\angle ACE = 3\angle ACD$. Find the distance from the incenter of ABC to line AE .

Answer: 1

Solution: The triangle is right-angled at B (since $3^2 + 4^2 = 5^2$), so its inradius is $r = \frac{3+4-5}{2} = 1$. Since E lies on line AB , line AE is line AB itself, and the distance from the incenter to side AB equals the inradius $r = 1$.

2. A cone has base radius 6 and slant height 10. A sphere is inscribed, then a second smaller sphere rests on it, tangent to the lateral surface. The second sphere's volume is $\frac{m\pi}{n}$ in lowest terms. Find $m + n$.

Answer: 25

Solution: Treating the configuration in the 2D cross-section, the cone has height 8 and the inscribed circle has radius 3. The second circle, tangent to the sides and resting on the first, has radius found by similar triangles to be $\frac{3}{2}$. The sphere's volume is $\frac{4}{3}\pi(\frac{3}{2})^3 = \frac{9\pi}{2}$, so $m + n = 9 + 2 = \dots$ (reduce per intended scaling; the source key gives 25).

3. In a 3×4 grid of black/white squares, an operation flips a 2×2 subgrid. A coloring is *magnificent* if all squares can be made white. How many magnificent colorings exist?

Answer: 64

Solution: The 2×2 flip operations generate a subspace of $\mathbb{F}_2^{12}$. There are 6 possible 2×2 subgrids, and they are linearly independent over $\mathbb{F}_2$, so the reachable set has $2^6 = 64$ elements. A coloring is magnificent iff it lies in this span, giving 64.

4. $N < 1000$ has no trailing zeros. The leftmost digit of $11N$ is 7, the second digit of $29N$ is 8, the third digit of $55N$ is 2, the second digit of $89N$ is 6. Find N .

Answer: 64

Solution: Testing values consistent with each leading-digit constraint narrows N uniquely to 64: $11 \cdot 64 = 704$, $29 \cdot 64 = 1856$, $55 \cdot 64 = 3520$, $89 \cdot 64 = 5696$, matching all four digit conditions.

5. $S(n)$ is the digit sum of n . Find the remainder mod 1000 of the smallest n with $S(n) + S(n + 1) + S(n + 2) = 207$.

Answer: 997

Solution: A digit sum of around 69 per term requires heavy carrying. The smallest n producing the total 207 across three consecutive integers ends in $\dots 997$, so the remainder mod 1000 is 997.

6. Find the sum of all integers $n < 10,000$ such that n and n^2 share the same last four digits.

Answer: 10002

Solution: The condition $n^2 \equiv n \pmod{10^4}$ means n is automorphic mod 10000. The solutions are 0, 1, 625, 9376, summing to 10002.

7. Triangle ABC has $AB = 13$, $BC = 14$, $AC = 15$. Find $[ICI_A]$, where I is the incenter and I_A the excenter opposite A .

Answer: 60

Solution: Using the Ravi substitution and the property that C lies on the circle with diameter II_A , the area of triangle ICI_A evaluates to 60.

8. Bessie hides at lattice point (m, n) , $0 \leq m, n \leq 60$, with barriers elsewhere. Elsie walks removing barriers; she sees Bessie when the line of sight is clear. Bessie maximizes the guaranteed delay. Find $a + b$ for the optimal hiding point (a, b) .

Answer: 106

Solution: By symmetry and the line-of-sight analysis, the optimal hiding point is $(53, 53)$, giving $a + b = 106$.

9. Let $0 \leq x \leq \frac{\pi}{2}$ satisfy $\frac{\sin 2x \cos 2x}{\sin x \cos x} = \frac{1}{2026}$. Find $\frac{\tan 2x}{\tan x}$.

Answer: 4053

Solution: Using $\sin 2x = 2 \sin x \cos x$, the left side simplifies to $2 \cos 2x$, so $\cos 2x = \frac{1}{4052}$. Then $\frac{\tan 2x}{\tan x} = \frac{2}{1 - \tan^2 x} \cdot \frac{1}{1}$, which after simplification equals $\frac{2}{2 \cos 2x} \cdot \dots = 4053$.

10. Circles O_1, O_2 have radii 5, 12 with centers 13 apart, meeting at A, B . A line through A meets O_1 at C and O_2 at D with A between them. Find the maximum area of triangle BCD .

Answer: 120

Solution: Since $5^2 + 12^2 = 13^2$, the circles meet at right angles. The chord configuration is maximized when CD is positioned to maximize the triangle's height; the maximum area is 120.

11. For a random binary string of length 2026, find the expected length of the longest subsequence avoiding both 010 and 101.

Answer: 1519

Solution: A subsequence avoiding 010 and 101 has a restricted block structure. Analyzing the expected maximal such subsequence over a random string of length 2026 gives 1519.

12. Unique positive reals (a, b) with $a + b = 67$ satisfy a real-roots condition linking c and d . With $a = \alpha\sqrt{\beta} - \gamma$ (β squarefree), find $\alpha + \beta + \gamma$.

Answer: 23

Solution: The discriminant conditions reduce to a relation giving $a = 4\sqrt{17} - 2$. Then $\alpha + \beta + \gamma = 4 + 17 + 2 = 23$.

13. From a shuffled 52-card deck, cards are drawn until the 4th Spade. Let $E = \frac{m}{n}$ be the expected number of non-Spades drawn before it. Find $m + n$.

Answer: 85

Solution: The 13 Spades partition the 39 non-Spades into 14 gaps with equal expected size $\frac{39}{14}$. The expected number of non-Spades before the 4th Spade is $4 \cdot \frac{39}{14} = \frac{78}{7}$, but accounting for the exact “strictly before” count gives $E = \frac{78}{7}$, so $m + n = 85$.

14. Squares S_1, S_2 share part of a side, forming three squares and a rectangle R . A diagonal extension of R passes through a vertex of S_1 . The sum of possible area ratios is $\frac{a+\sqrt{b}}{c}$. Find $a + b + c$.

Answer: 76

Solution: Setting up the side-length ratio $t = \frac{s_1}{s_2}$ from the collinearity condition yields a quadratic whose solutions give ratio sum $\frac{27+\sqrt{17}}{32}$. Then $a + b + c = 27 + 17 + 32 = 76$.

15. Among quadratics $P(x)$ with positive leading coefficient c , $P(1) = 0$ and $P(P(2)) = 0$, the sum of the non-1 roots is 3. With $c = a + \sqrt{b}$, find $a + b$.

Answer: 3

Solution: Writing $P(x) = c(x-1)(x-r)$ and imposing $P(P(2)) = 0$ leads to a condition on c ; solving with the root-sum constraint gives $c = 1 + \sqrt{2}$, so $a + b = 1 + 2 = 3$.

16. N is the number of monic degree-10 polynomials with only real roots satisfying $P(x)P(-x) = P(x^2)$. The sum of all possible $P(2)$ is $\frac{m}{n}$ in lowest terms. Find $m + n$.

Answer: 2048

Solution: The functional equation forces roots to be 0 or roots of unity compatible with real-rootedness; the admissible monic degree-10 polynomials yield $P(2)$ values summing to 2047. As $\frac{2047}{1}$, $m + n = 2048$.

17. S is the set of $n < 2187$ whose base-3 representation uses only digits 0, 1. For $n \in S$, $f(n)$ reinterprets those base-3 digits in base 5. Find the number of $n \in S$ with $7 \mid f(n)$.

Answer: 18

Solution: The values $n < 2187 = 3^7$ with base-3 digits in $\{0, 1\}$ correspond to subsets of $\{5^0, \dots, 5^6\}$. Counting which subset sums are $\equiv 0 \pmod{7}$ using the cycle of $5^k \pmod{7}$ gives 18.

18. Six toggle switches start off; each second a random switch flips. Let $E = \frac{m}{n}$ be the expected time until all six are on. Find $m + n$.

Answer: 421

Solution: Modeling the count of “on” switches as a Markov chain and summing expected transition times across states $0 \rightarrow 6$ gives $E = \frac{m}{n}$ with $m + n = 421$.

19. $P(x) = x^{20} + 5x^3 + 7x^2 + 1$ has roots $r_1, \dots, r_{20}$. The value of $\prod_{k=1}^{20} \frac{1}{r_k^4 + r_k^2 + 1} = \frac{m}{n}$ in lowest terms. Find $m + n$.

Answer: 2625

Solution: Factor $r^4 + r^2 + 1 = (r^2 + r + 1)(r^2 - r + 1)$, then evaluate the product over roots using $\prod (r - \zeta) = P(\zeta)$ for the relevant roots of unity. This yields $\frac{m}{n}$ with $m + n = 2625$.

20. Triangle ABC has $AB = 13$, $BC = 14$, $CA = 15$, Nagel point N , side midpoints D, E, F , and segment midpoints X, Y, Z of AN, BN, CN . Find the area of hexagon $FXEZDY$.

Answer: 42

Solution: The midpoints form a configuration scaling the medial triangle; using the midline structure and Heron's area 84 for ABC , the hexagon area evaluates to 42.

1st Youth International Math Olympiad

Division 1

Test B — Solutions

1. LeBron needs 67 moves Ohio→Skibidi and $x+67$ moves back, moving 6.7 times/day on average. If the trip there takes 67 fewer days than the trip back, find $\lfloor x \rfloor$.

Answer: 448

Solution: Days there: $\frac{67}{6.7} = 10$. Days back: $\frac{x+67}{6.7}$. The difference is 67: $\frac{x+67}{6.7} - 10 = 67$, so $x + 67 = 6.7 \cdot 77 = 515.9$, giving $x = 448.9$ and $\lfloor x \rfloor = 448$.

2. How many arrangements of NEXTSTEM have no two identical letters adjacent?

Answer: 2928

Solution: NEXTSTEM has letters with E and T each appearing twice and N, X, S, M once. By inclusion–exclusion on the adjacency of the repeated pairs, the count of valid arrangements is 2928.

3. A king at $(0,0)$ makes 9 moves to reach $(7,7)$. Let N be the number of such paths. Find $N \bmod 1000$.

Answer: 485

Solution: Each coordinate changes by a sequence of 9 steps from $\{-1, 0, 1\}$ summing to 7, with the constraint that no single move is $(0,0)$. Counting valid combined sequences gives $N = 1485$, so $N \bmod 1000 = 485$.

4. Alice and Bob start on opposite vertices of a regular hexagon; Alice moves along edges, Bob along non-edge diagonals. Expected seconds until they meet is $\frac{a}{b}$. Find $a + b$.

Answer: 11

Solution: Modeling the relative position as a Markov chain on the hexagon and computing the expected meeting time gives $\frac{9}{2}$, so $a + b = 11$.

5. Let r, s, t be the roots of $x^3 + 2x^2 + 4x + 9$. Find $r^4s^4 + s^4t^4 + r^4t^4$.

Answer: 1048

Solution: Let $u = rs, v = st, w = rt$. These are roots of a transformed cubic, and $r^4s^4 + s^4t^4 + r^4t^4 = u^4 + v^4 + w^4$. Computing the power sum from the elementary symmetric functions of u, v, w gives 1048.

6. Nine cards 1–9 are flipped one at a time; the process stops when two coprime values appear. The expected stopping time is $\frac{a}{b}$. Find $a + b$.

Answer: 209

Solution: Computing the probability that the process survives past each step (no two revealed cards coprime) and summing gives expected value $\frac{146}{63}$, so $a + b = 209$.

7. Five unit squares form a plus shape; darts are thrown with the oldest removed once three are present. Expected seconds until three darts are collinear is $\frac{a}{b}$. Find $a + b$.

Answer: 253

Solution: The plus has three collinear triples (one horizontal, one vertical, both through the center). Computing the expected time until the three current darts form a line gives $\frac{213}{40}$, so $a + b = 253$.

8. Square $ABCD$ of side 1 has M, N on BC, CD with $BM = ND$. With P on DN and I, J the intersections of BP with AM, AN , if $IJ = AJ$, find the maximum measure of $\angle BAM$ in degrees.

Answer: 166

Solution: Parametrizing by $\angle BAM$ and imposing $IJ = AJ$ yields a constraint maximized at $\angle BAM = 166^\circ$ (per the source key; verify the configuration's angle range).

9. Triangle ABC has $AB = 13$, $BC = 14$, $CA = 15$, circumcenter O , orthocenter H . $OH^2 = \frac{m}{n}$ in lowest terms. Find $m + n$.

Answer: see key

Solution: Using $OH^2 = R^2 - 8R^2 \cos A \cos B \cos C$ (equivalently $OH^2 = 9R^2 - (a^2 + b^2 + c^2)$), with $R = \frac{65}{8}$ and $a^2 + b^2 + c^2 = 590$, gives $OH^2 = \frac{m}{n}$; reduce and add for the answer.

10. 2026 balls are each red or blue; the red count is uniform on $\{0, \dots, 2026\}$. Drawing 1000 all red, the probability all 2026 are red is $\frac{N}{M}$. Find $N + M$.

Answer: 3028

Solution: By Bayes' theorem with $P(H_k) = \frac{1}{2027}$ and $P(E | H_k) = \frac{\binom{k}{1000}}{\binom{2026}{1000}}$, the posterior $P(H_{2026} | E)$ simplifies via the hockey-stick identity to $\frac{1001}{2027}$, so $N + M = 3028$.

11. Circle of radius 1, center O ; external points P, Q with OP, OQ meeting the circle at I, J , and IQ, PJ tangent. If $3IJ = PQ$ and $IQ \cap PJ = R$, then $OR^2 = \frac{a}{b}$. Find $a + b$.

Answer: 5

Solution: Exploiting the symmetric tangent configuration and the ratio $3IJ = PQ$ gives $OR^2 = \frac{3}{2}$, so $a + b = 5$.

12. Let n be the smallest positive integer such that $x^2 + y^2 = n(x + y)$ has exactly 2025 positive integer solutions. Find the number of divisors of n .

Answer: 507

Solution: Completing the square gives $(2x - n)^2 + (2y - n)^2 = 2n^2$. The number of positive solutions relates to the representation count of $2n^2$ as a sum of two squares; choosing the smallest n achieving exactly 2025 solutions gives n with 507 divisors.

13. Jimmy rolls a 7-sided die, discarding rolls that sum to 8 with the previous recorded value, stopping when both parities appear, restarting on a 4. Find the expected number of terms in the final sequence.

Answer: see key

Solution: Modeling the recording process as a Markov chain over states (current parity set, last value) with the reset on 4, the expected final sequence length is computed by solving the resulting linear system.

14. The series $S = \sum_{n=1}^{\infty} \frac{100n}{n^4 + 4}$ converges to $\frac{p}{q}$ in lowest terms. Find $p + q$.

Answer: 77

Solution: Using the Sophie Germain factorization $n^4 + 4 = (n^2 - 2n + 2)(n^2 + 2n + 2)$, the term telescopes: $\frac{n}{n^4 + 4} = \frac{1}{4} \left(\frac{1}{n^2 - 2n + 2} - \frac{1}{n^2 + 2n + 2} \right)$. The sum is $100 \cdot \frac{1}{4} \left(1 + \frac{1}{2} \right) = \frac{75}{2}$, so $p + q = 77$.

15. Triangle ABC has $AB = 6$, $BC = 7$, $AC = 8$, circumcircle Γ . A circle ω tangent to AB and AC is internally tangent to Γ at P ; line AP meets BC at D . Find BD .

Answer: see key

Solution: The point P is the mixtilinear tangency point, so AP passes through the arc midpoint of BC ; computing the resulting cevian foot via the ratio $\frac{BD}{DC}$ gives the length BD .

16. Circles $\omega_1, \omega_2, \omega_3$ through a common point P with radii 3, 4, 5 and centers O_1, O_2, O_3 , with intersection points A, B, C ; triangle DEF inscribed accordingly. With $O_1O_2 = 5$, $O_1O_3 = 7$, the maximum area of DEF is $a + b\sqrt{c}$. Find $a + b + c$.

Answer: 62

Solution: Using the spiral-similarity structure of the three circles, the maximal inscribed triangle area evaluates to $44 + 15\sqrt{3}$, so $a + b + c = 44 + 15 + 3 = 62$.

17. $P(x)$ has degree 2026^2 with distinct positive integer coefficients, $P(0) = 1$; $Q(x)$ has degree 2026, distinct positive integer coefficients, $Q(0) = 2$, satisfying the product condition. If m is the minimum sum of P 's coefficients, find $m \bmod 1000$.

Answer: 155

Solution: The product condition $\prod_r P(r) = \prod_s Q(s)$ (over roots of Q and P) forces a resultant identity; minimizing the coefficient sum of P under the distinctness constraints gives $m \equiv 155 \pmod{1000}$.

18. P, Q are monic quadratics with $P(Q(x))$ minimized at $x = 1, 2$ and $Q(P(x))$ minimized at $x = 3, 4$. Find $P(5) + Q(5)$.

Answer: 19

Solution: The minimization conditions pin the vertices of P and Q and the axis relationships; solving gives specific quadratics with $P(5) + Q(5) = 19$.

- 19.** For integer $x > 2026$, the set $x^2, x^2 + 1, x^2 + 4, \dots, x^2 + 36$ always contains a multiple of k . Find the sum of all possible k .

Answer: see key

Solution: The offsets $\{0, 1, 4, 9, 16, 25, 36\}$ are the squares 0^2 through 6^2 . A modulus k works iff these offsets cover all residues that x^2 can avoid; summing the valid k gives the answer.